0 Utility

0.1 Common Unit Conversions

- 1 atm = 33.9 ft wg
- 1 gal = $\frac{1}{7.48}$ ft^3
- 1 gpm = $\frac{1}{448.8}$ ft^3/s
- 1 cfm = 0.0004719 m^3/s
- $g = 9.81$ m/s^2 = 32.2 ft/s^2
- 1 J/kg = 1 m^2/s^2
- 1 psi = 6895 Pa
- 1 in. wg = 249.1 Pa at 70°F
- 1 psi = 27.68 in. wg
- 1 ft = 12 in, 1 m = 3.281 ft

2 Air Duct Systems

2.1 General Procedure

1. **Assumptions:**
   - Maximum velocity
     - Non-industrial: maximum velocity = **1200 fpm**
     - Industrial: maximum velocity = **2200 fpm**
     - If CFM is capped, check **Table 2.3** for maximum velocity.
   - If P_{fan} unknown, target pressure drop: **0.1 inch wg per 100 ft of duct**[1]. Else, when pressure drop is known, calculate friction loss:
   $$\text{friction loss} = \frac{P_{\text{fan}} - P_{\text{longest exit}}}{L_{\text{eq}}} \cdot 100 \text{ ft}$$
   - Use the **equal friction method** to size and balance the system.
     - Aim to have equal friction loss in all branch sections of the duct system.
     - Use dampers to equalize ΔP across branches, if needed.
   - **Material selection:**
     - **Sheet Metal**
       * *Galvanized steel*: Typical for air duct system; Good corrosion resistance; cost-effective.
       * *Aluminum*: Lightweight, corrosion-resistant; easier handling.
       * *Stainless steel*: High corrosion resistance; suitable for high pressure and temperature.
     - **PVC (Plastic)**
       * *ASTM D 2513*: Thermoplastic gas pressure pipe (polyethylene), for temperatures > 428°F.
       * *ASTM D 2517*: Reinforced epoxy resin pipe; very high corrosion resistance, pressures > 8300 psi.
   - **Component Selection**
     - *Bellmouth:* Used at entrance to reduce noise.
     - *Tee 90°:* Used to split flow into two ducts.
     - *Elbow 90°:* Used to change direction of flow.
     - *Diffuser:* To distribute air. Check **Table 2.2** for pressure loss.

2. **Pipe Sizing**
   - Use **Fig A.1** To find what you don't have. You need three of cfm, friction loss, duct size, and velocity to find the last one. Usually use cfm and friction with velocity maximums to find duct size.
   - Choose **even** size if reasonable because they are more commonly available.
   - Write down the diameter, cfm, friction loss in a nice table.

3. **Calculate Pressure Losses**
   Check the main branch and then offshoots from the main branch to determine highest pressure loss.
   - Calculate pressure loss due to friction:
   $$\Delta P_{i-j} = (\text{friction loss}) \cdot L_{\text{eq, i-j}} \quad [\text{in. wg}]$$
     - $L_{\text{eq, i-j}}$ = equivalent length of duct section between points i and j which can be caluclated by summing length of pipe and adding equivalent lengths of components from **Table A.4**.
   - Calculate pressure loss due to components by referring to **Table 2.2**.
   - Total pressure loss:
   $$\Delta P = \Delta P_i + \sum \Delta P_{\text{components}} \quad [\text{in. wg}]$$

4. **Choose a Fan:**
   - Select a fan that meets the required pressure drop and flow rate.
   - Ensure the fan is suitable for the application (e.g., centrifugal, axial).

5. **Draw final layout and balance system:**
   - Label each section with flow rate, duct size, length, and pressure drop.
   - Add dampers and VFDs as needed for balancing flow across branches.

6. **Conclusion**
   - Ensure all components are properly sized and selected.
   - Verify that the system meets design criteria for flow rate, pressure drop, and noise levels.
   - Discuss improvements

[1] Most common case.

2.2 Reynolds Number Concepts

2.2.1 Internal Flow and Flow Regimes

- **Laminar flow:** Smooth, ordered motion.
- **Turbulent flow:** Disordered, fluctuating motion.
- Characterized by the Reynolds number:
$$\mathrm{Re} = \frac{\rho V D_h}{\mu} \quad \text{or} \quad \mathrm{Re} = \frac{V D_h}{\nu}$$
- For non-circular ducts, use hydraulic diameter:
$$D_h = \frac{4A_c}{P}$$
(where A_c is cross-sectional area, P is perimeter)

Hydraulic Diameters for Common Duct Shapes:

- Square duct ($L \times L$):
$$D_h = \frac{4(L \cdot L)}{4L} = L$$
- Rectangular duct ($L \times w$):
$$D_h = \frac{4Lw}{2L + 2w} = \frac{2Lw}{L + w}$$
- Annular duct (between outer D_2 and inner D_1):
$$D_h = \frac{D_2^2 - D_1^2}{D_2 + D_1} = D_2 - D_1$$

Note: For flow rate, use actual area not D_h.

2.2.2 Flow Characterization

- Use critical Reynolds number: $\mathrm{Re}_{\mathrm{cr}} = 2300$
- $\mathrm{Re} < \mathrm{Re}_{\mathrm{cr}}$: Laminar flow
- $\mathrm{Re} > \mathrm{Re}_{\mathrm{cr}}$: Turbulent flow

2.3 Head Losses

2.3.1 Major and Minor Losses

- Major losses (H_l): Caused by wall friction and viscous effects in fully developed flow through straight duct sections.
- Minor losses (H_{lm}): Result from components such as entrances, exits, bends, fittings, valves, and sudden area changes. In air ducts, this includes filters and heating/cooling coils.
- **Units:** h_l, h_{lm} in J/kg or m^2/s^2 (energy head); H_l, H_{lm} in m or ft (elevation head)
- Total head loss:
$$H_{lT} = H_l + H_{lm} = \frac{h_l + h_{lm}}{g}$$

2.3.2 Major Head Loss Formula

$$h_l = f \cdot \frac{L}{D} \cdot \frac{V^2}{2} \qquad \text{or} \qquad H_l = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

- For **laminar flow**:
$$f = \frac{64}{\mathrm{Re}}$$
- For **turbulent flow**, use the Moody chart (default method):
  1. Calculate Re
  2. Determine relative roughness ε/D (from Table A-1)
  3. Use Re and ε/D to find f on the chart
  4. Compute head loss using f
- Alternatively, use:
$$\frac{1}{\sqrt{f}} = -1.8 \log_{10}\left[\frac{6.9}{\mathrm{Re}} + \left(\frac{\varepsilon/D}{3.7}\right)^{1.11}\right] \quad \text{(Haaland)}$$

2.3.3 Minor Head Loss Formula

$$h_{lm} = K \cdot \frac{V^2}{2} \qquad \text{or} \qquad H_{lm} = K \cdot \frac{V^2}{2g}$$
$$h_{lm} = f \cdot \frac{L_{\mathrm{equiv}}}{D} \cdot \frac{V^2}{2}$$

- K = loss coefficient (from tables)
- L_{equiv} = equivalent length of fitting/device

2.4 Rectangular Duct Sizing and Aspect Ratio

- Use hydraulic diameter for Reynolds number and friction calculations:
$$D_h = \frac{2Lw}{L + w}$$
- Cross-sectional area:
$$A = L \cdot w$$
- **Aspect Ratio (AR)**:
$$\mathrm{AR} = \frac{L}{w} \quad \text{(smaller is better)}$$
  - Recommended: $\mathrm{AR} < 4$
  - If $\mathrm{AR} > 4$:
    - Higher frictional losses (more perimeter for same area)
    - More expensive fabrication
- **Design Steps:**
  1. Use circular equivalent method to determine round duct size based on Q and acceptable V (see Fig. A.1)
  2. Select a rectangular duct with same area and acceptable aspect ratio (see Table A.3)
  3. Check space, noise, and architectural constraints

3 Liquid Pipe System

1. **Define system and assumptions:**
   - Identify known values (e.g., Q, pipe dimensions, velocity, elevation changes).
   - Set fluid properties (e.g., ρ, μ at given temperature). **Table HEX liquid-gas on Canvas**.
   - List all fittings, joints, valves with K values or equivalent lengths.
     - **Flanged fittings** are popular
   - **Schedule 40 steel** is popular (low carbon steel).
   - Typical velocities can be found in **Table A.12**.
     - **Cold water supply piping**: 4-8 ft/s
     - **Pump suction line**: ≤ 5 ft/s
   - Check erroosion limits in **Table A.13**.
   - Target pressure drop in pipting: $\leq$ **3 ft wg/100 ft**
   - Recommended to convert gpm to ft^3/s for calculations. 1 gpm = 1/448.8 ft^3/s
2. **Sizing**
   - If pipe diameter is not given, check **Fig. A.3** or **Fig. A.4**, depending on material.
     - **Open system** if the system is open to the atmosphere.
     - **Closed system** if the system is closed to the atmosphere.
3. **Apply the energy equation:**

$$H_{\text{pump}} = \frac{w_{\text{pump}}}{g} = H_{lT} + \left[\frac{P_2}{\rho g} + \alpha_2 \frac{V_{\text{ave},2}^2}{2g} + z_2\right] - \left[\frac{P_1}{\rho g} + \alpha_1 \frac{V_{\text{ave},1}^2}{2g} + z_1\right]$$

   $\alpha = 2$ for laminar flow, $\alpha = 1.06 \approx 1$ for turbulent flow
4. If ① and ② are:
   - Both upstream or both downstream of the pump: $H_{\text{pump}} = 0$
   - On opposite sides of the pump: $H_{\text{pump}} \neq 0$
5. **Compute head losses:**
   - **Major loss:**

$$H_{\text{major}} = f \cdot \frac{L}{D} \cdot \frac{V^2}{2g}$$

     - Find f using the Moody chart with ε/D (see **Table A.1**), or use the Swamee–Jain equation:

$$f = 0.25 \left[\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{5.74}{\text{Re}^{0.9}}\right)\right]^{-2}$$

     - Other friction factor formulas are listed in **Table A.2**.
     - **If Using Sizing Chart:**

$$H_l = \frac{(\text{loss per 100 ft})}{100} \cdot L_{\text{pipe}} \quad [\text{ft}]$$

       * Used when referencing **Table A.3** or **Table A.4** for direct friction loss values.
   - **Minor loss:**

$$H_{\text{minor}} = \sum K \cdot \frac{V^2}{2g}$$

     - Use K values from **Table A.14**.
     - Entrance losses: $K_{\text{inlet}} = 0.78$ (from notes); sometimes $K_{\text{inlet}} = 0.5$ and $K_{\text{outlet}} = 1$ may be used.
   - **Total head loss:**

$$H_{lT} = H_{\text{major}} + H_{\text{minor}}$$

6. **Determine flow parameters:**
   - Compute V_{ave}:

$$V_{\text{ave}} = \frac{Q}{A} = \frac{4Q}{\pi D^2}$$

   - Compute Reynolds number:

$$\text{Re} = \frac{4\rho Q}{\pi \mu D} = \frac{V_{\text{ave}} \rho D}{\mu}$$

   - Estimate friction factor f:
     - Use Moody chart if ε/D is known.
     - Or apply Haaland equation:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10}\left[\frac{6.9}{\text{Re}} + \left(\frac{\varepsilon/D}{3.7}\right)^{1.11}\right]$$

7. **Compute pump power:**

$$\text{bhp} = \frac{\rho g Q H_{\text{pump}}}{\eta_{\text{pump}}} \quad \Rightarrow \quad \text{Convert to hp as needed}$$

   or efficiency:

$$\eta_{\text{pump}} = \frac{\rho Q g H}{\omega T_{\text{shaft}}} = \frac{W_{\text{water horsepower}}}{\text{bhp}} = \frac{\rho Q g H}{\text{bhp}} = \frac{m g H}{\text{bhp}}$$

8. **Cavitation Check (NPSH):**
   - Use energy equation to find pressure at pump inlet:

$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_{\text{ave},1}^2}{2g} + z_1 = \frac{P_2}{\rho g} + \alpha_2 \frac{V_{\text{ave},2}^2}{2g} + z_2 + H_{lT,\text{suction}}$$

     Find $H_{lT,\text{suction}}$ using the same method as above.

- Compute:

  $$\text{NPSHA} = \left(\frac{P}{\rho g} + \frac{V_{\text{avg}}^2}{2g}\right)_{\text{pump inlet}} - \frac{P_{\text{vapor}}}{\rho g}$$

  vapor pressure can be found in **Table A-9E**.

- Compare with NPSHR to ensure:

  $$\text{NPSHA} > \text{NPSHR}$$

4 Heat Exchangers

4.1 General Procedure for Finned-Tube Heat Exchanger Design

1. **Define design goals and assumptions:**
   - Specify target heat transfer rate (e.g., 96,000 Btu/hr), air inlet/outlet temperatures, and water conditions.
   - Choose a reasonable air velocity over the coil. Common practice:
     - V_{air} = 400-500 fpm to reduce noise and align with filter performance.
     - 500 fpm is suitable for residential heating coils. Generally same max air velocity as air duct design (see Section 2).
   - Choose a water velocity in tubes:
     - $V_{\text{water}} \approx 4$ fps is typical for potable or building service water.
     - Below copper erosion limits and compatible with pump suction velocity constraints.
   - Use typical material choices:
     - Tube material: **Type K copper** (high conductivity, standard availability).
     - Fin material: **Copper** for improved heat transfer.
   - Select standard tube geometry:
     - Outer diameter: $D_o = 0.402$ in (approx. 3/8 in. nominal), same as **Table C.4a**.
     - Inner diameter: $D_i = 0.305$ in (based on Type K 1/4 in NPS, Table A.6).
     - Tube pitch: $x_L = 0.866$ in (longitudinal), $x_t = 1.0$ in (transverse) [Figure C.4a].
   - Use a **counterflow** configuration:
     - Higher effectiveness compared to parallel flow.
     - Staggered tube arrangement increases turbulence and enhances heat transfer.
2. **Properties of Fluid**
   Go to **Table A-15E** for air and **Table A-9E** for water properties. Note down the following:
   - Water properties: T_w, $T_{w,in}$, $T_{w,out}$, $c_{p,w}$, ρ_w, μ_w, k, Pr
   - Air properties: T_a, $T_{a,in}$, $T_{a,out}$, $c_{p,a}$, ρ_a, μ_a, k, Pr
3. **Calculate Reynolds Number:**
   $$\text{Re}_D = \frac{\rho_w V_w D_i}{\mu_w} \quad \text{(water side)}$$
4. **Determine Convective Heat Transfer Coefficients:**
   - Calculate $\bar{h}_i$ using the Dittus–Boelter equation (for water inside smooth tubes):
     $$\text{Nu} = 0.023 \cdot \text{Re}^{0.8} \cdot \text{Pr}^{0.3}$$
     $$\bar{h}_i = \frac{\text{Nu} \cdot k}{D} \quad \left[\frac{\text{Btu}}{\text{hr} \cdot \text{ft}^2 \cdot {}^\circ\text{F}}\right]$$
     Valid if:
     - Flow is fully developed and turbulent
     - Tubes are smooth
     - Re > 10,000
     - 0.7 < Pr < 160
     - $L/D_{\text{inner}} > 60$
   - Calculate $\bar{h}_o$ from j-factor charts (air side).
     (a) Pull up **Figure C.4a**. Note down, D_h, σ, thin thickness.
     (b) Calculate G_m:
     $$G_m = \frac{\rho_{\text{air}} V_{\text{air}}}{\sigma} \quad [\text{lb/s-ft}^2]$$
     (c) Calculate Re_{G_m} using D_h from chart:
     $$\text{Re}_{G_m} = \frac{G_m D_h}{\mu_{\text{air}}} \quad \text{[dimensionless]}$$
     (d) Find j-factor from **Figure C.4a**.
     (e) Calculate h_o:
     $$\bar{h}_o = \frac{j \cdot G_m c_{p,\text{air}}}{\text{Pr}^{2/3}} \quad [\text{Btu/hr-ft}^2\text{-F}]$$
5. **Fouling Resistance**
   Can be found in **Table C.3**
6. **Fin Efficiency**
   i. Fin surface effectiveness:
   $$\eta_{\text{so}} = 1 - \frac{A_{\text{fin}}}{A}(1 - \eta_{\text{o}})$$
   ii. Fin efficiency:
   $$\eta_o = \frac{\tanh(mr\phi)}{mr\phi}$$
   iii. Find m. From j-factor chart find fin thickness, t_{fin}. Then calculate:
   $$m = \frac{(2h_o)^{1/2}}{(k_{\text{fin}} t_{\text{fin}})^{1/2}} \quad [\text{ft}^{-1}]$$
   iv. Find r by
   $$r = \frac{D_{\text{tube}}}{2}$$
   v. Find ϕ by
   $$\phi = \left(\frac{R_{\text{EQ}}}{r} - 1\right)\left[1 + 0.35 \ln\left(\frac{R_{\text{EQ}}}{r}\right)\right]$$

vi. Find R_{EQ}/r to use in ϕ:
$$R_{\text{EQ}} = 1.27\psi(\beta - 0.3)^{1/2}$$

vii. Find ψ from numbers from the j-factor chart:
$$\psi = \frac{x_t}{D_o}$$
where x_t is the transverse tube pitch (1.00 in) and D_o is the outer diameter of the tube.

viii. Find β from the j-factor chart:
$$\beta = \frac{L}{M}$$

ix. Find M to use in β:
$$M = \frac{x_t}{2}$$

x. Find L to use in β:

xi.
$$L = \frac{\left[M^2 + x_L^2\right]^{1/2}}{2}$$
where x_L is the longitudinal tube pitch (0.866 in).

xii. Shove the numbers to solve for η_o and η_{so}.

7. A_i/A_o **Ratio**
$$\frac{A_i}{A_o} \approx \left(\frac{A_i}{\Omega} \bigg/ \frac{A_o}{\Omega}\right) \approx \frac{\pi D_i}{x_t x_L \alpha} = \frac{\pi D_i}{\alpha x_t x_L}$$

8. **Calculate Overall Heat Transfer Coefficient:**
   - Use:
$$U = \frac{A_o}{A_i}\left(R_{fi} + \frac{1}{h_i}\right) + \frac{1}{\eta_o h_o} + R_{fo}$$
   - Where:
     - R_{fo} = fouling resistance on the outside of the tube (ft^2-F/Btu) (air)
     - R_{fi} = fouling resistance on the inside of the tube (ft^2-F/Btu) (water)

9. **Use ε-NTU method to find surface area:**
   - Find total surface area:
$$A_o = \frac{C_{\min}\text{NTU}}{U}$$
   - Find C_{air} using
$$\dot{Q} = C_{\text{air}}\left(T_{a,out} - T_{a,in}\right)$$
   - Find C_{water} using
$$\dot{Q} = C_{\text{water}}\left(T_{w,out} - T_{w,in}\right)$$
   - Find $C_{\min}$ by taking the minimum of the two. If $C_{\min}/C_{\max} \approx c_{p,a}/c_{p,w}$ then say the assumption of $\dot{m}_a = \dot{m}_w$ is valid.
   - Find ε:
   -
$$\varepsilon = \frac{C_{\text{cold}}}{C_{\min}} \frac{T_{\text{cold},out} - T_{\text{cold},in}}{T_{\text{hot},in} - T_{\text{cold},in}}$$
   - Use **Figure 4.12** to find NTU
   - Use **Table 4.5** to find A_o.

10. **Find Number of Rows and Tubes:**
    - Use:
$$\Omega = \frac{A_o}{\alpha}, \quad W = \frac{\Omega}{A_f}$$
Where A_f is the frontal area of the heat exchanger.
    - Number of tube rows:
$$N_r = \frac{W}{x_L}, \quad \text{round up}$$
Where x_L is the longitudinal tube pitch (0.866 in). N_r is the number of rows of tubes in the heat exchanger.
    - Number of tubes per row:
$$\dot{m}_w = \rho_w V_w A_{\text{tube}} = N_{\text{tube}} \cdot \frac{\pi D_i^2}{4}$$
and
$$C_{\text{hot}} = \dot{m}_w c_{p,w} = \frac{\dot{Q}}{\Delta T_w} \quad \text{(water side)}$$
Rearranging gives:
$$N_{\text{tube}} = \frac{4C_{\text{hot}}}{c_{p,w}\rho_w V_w \pi D_i^2}$$
Note convert inches to feet and hours to seconds.

11. **Size of the heat exchanger:**
    - Height:
$$H = N_{\text{tube}} \cdot x_t$$
    - Width:
$$W = \frac{A_f}{H}$$

12. **Look at Problem set 3 for the rest of the calculations.**